

Computational Mathematics

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1	Numerical Analysis	1
2	Numerical Optimization	3
3	Numerical Linear Algebra	5
4	Optimization on Manifolds	6
5	Numerical Partial Differential Equations	7
6	Tensor Decompositions and Their Applications	8
7	Fast Algorithms and Integral Equation Methods	9
8	Compressed Sensing and Sparse Signal Processing	10
9	Solving Partial Differential Equations in Parallel on GPUs	11

1 Numerical Analysis

1. Class introduction, absolute and relative errors, and vector norms
2. Forward/backward errors, condition numbers, stability, accuracy, and vector norm equivalence.
3. Floating-point basics, subnormal numbers, and rounding methods
4. Linear systems, matrix norms, conditioning, residuals, perturbations, and preconditioning
5. SVD, simple linear systems, LU factorization, pivoting, and Cholesk
6. LU applications, BLAS/LAPACK, Schur complements, special cases, round-off, Sherman-Morrison.
7. Least squares basics, normal equations, orthogonal projectors, and pseudoinverse methods
8. Least squares sensitivity, QR factorization, and QR computation methods
9. QR via Householder, Givens rotations, and rank-deficient least squares
10. SVD applications, reduced forms, eigenvalue basics, and eigenvalue transformations
11. Eigenvalue sensitivity, power iteration, Rayleigh quotient, and Schur form
12. Multiple eigenvalues, QR algorithm, forward/inverse methods, and computational cost
13. Krylov methods, Arnoldi iteration, SVD computation, nonlinear equations, and sensitivity
14. Convergence rates, bisection, Taylor's theorem, fixed-point iteration, and Newton's method
15. Secant, rootfinding methods, multidimensional fixed-point/Newton, Broyden, and derivative testing
16. Optimization basics, convexity, optimality, sensitivity, golden-section search, and Newton's method
17. Gradient descent, conjugate gradients, Nelder-Mead, Newton, and quasi-Newton optimization methods
18. Nonlinear least squares and constrained optimization methods
19. Interpolation basics, Lagrange interpolation, Newton interpolation, and fundamental concept
20. Orthogonal polynomials and interpolation error estimation methods
21. Splines, quadrature basics, interpolatory quadrature, and accuracy analysis

22. Gaussian and composite quadrature, numerical differentiation, and differentiation matrices
23. Finite differences, Taylor expansions, Richardson extrapolation, and initial value problems
24. IVP well-posedness, Euler' s method, accuracy, and stability analysis
25. IVP stiffness, Runge-Kutta methods, and boundary value problem basics
26. Boundary value problem well-posedness, numerical methods, and Galerkin approaches
27. Stationary iterative methods and conjugate gradients for sparse linear systems
28. Partial differential equations
29. Review session

2 Numerical Optimization

1. Introducing the Standard Mathematical Formulation of General Constrained Optimization Problems
2. Definitions, Minimizer Existence, Notation, and Major Classes of Optimization Problems
3. Convexity Tests, Convex Sets, Preserving Operations, and Standard Optimization Form
4. Semidefinite Programming, Convex Optimality, Lagrangian Functions, Weak Duality, LP Duals
5. Dual Decomposition, Strong Duality, Optimality Conditions, Perturbation, and Estimation Problems
6. Linear Least Squares, Ill-Posed Problems, Regularization, Statistical Derivation, and L1-Estimation
7. Gauss-Newton, Levenberg-Marquardt, Exact Newton Methods, and Local Convergence Rate Analysis
8. Newton-Type Methods and Local Contraction Theorems for General Newton Iterations
9. Affine Invariance, Local Newton Convergence, and Armijo Backtracking Line Search
10. Wolfe Conditions and Global Convergence of Armijo Line Search Methods
11. Trust-Region Methods, Cauchy Point Computation, and Trust-Region Step Selection
12. Algorithmic Differentiation, Forward and Backward Modes, and Derivative Computation Methods
13. Algorithmic Differentiation Software and Introduction to Equality-Constrained Optimization Conditions
14. LICQ, Linearized Feasible Cones, and Second-Order Necessary Optimality Conditions
15. Optimality Conditions, Equality-Constrained QP, KKT Systems, Newton-Lagrange, and Constrained Gauss-Newton
16. Perturbation Analysis and Local Convergence in Equality-Constrained Numerical Optimization Methods
17. Constrained Newton-Type Algorithms, Local Convergence, Line Search, and BFGS Updating
18. KKT Necessary Conditions, Active Constraints, Constraint Qualification, and LICQ

19. Convex Problems, LICQ, KKT Conditions, and Global Optimality Characterization Theorem
20. Complementarity, Second-Order Conditions, and Active-Set Methods for Quadratic Programming Problems
21. Sequential Quadratic Programming, Powell's SQP Algorithm, and Interior Point Methods
22. Barrier Problems, Interior-Point Methods, and Sequential and Simultaneous Optimal Control
23. Problem Reformulations and Useful Function Approximations for Efficient Numerical Optimization
24. Summary of the Course

3 Numerical Linear Algebra

1. [Introduction, Gaussian Elimination](#)
2. [LU Decomposition](#)
3. [Special Matrices](#)
4. [Norms; The Condition Number](#)
5. [The Full Rank Least Squares Problem](#)
6. [QR Factorization via Givens Rotations](#)
7. [QR Factorization via Householder Reflections](#)
8. [Gram-Schmidt Orthogonalization](#)
9. [Singular Value Decomposition](#)
10. [SVD cont'd, Least Squares with Linear Constraints](#)
11. [Method of Steepest Descent](#)
12. [Lanczos Iteration](#)
13. [The Conjugate Gradient Method](#)
14. [Other Krylov Subspace Methods](#)
15. [Eigenvalues: Definitions and Properties](#)
16. [Schur Decomposition, Jordan Canonical Form](#)
17. [Perturbation Theory; Power Iterations](#)
18. [Hessenberg QR](#)
19. [Shifted QR Iteration](#)
20. [The Symmetric Eigenvalue Problem](#)
21. [The SVD Algorithm](#)
22. [Introduction to Statistical Learning](#)
23. [Principal Component Analysis](#)

4 Optimization on Manifolds

5 Numerical Partial Differential Equations

6 Tensor Decompositions and Their Applications

1. [Introduction and an Example](#)
2. [CP Decomposition theory](#)
3. [CP Theory](#)
4. [CP Theory](#)
5. [CP Theory and Applications](#)
6. [CP Applications](#)
7. [CP Applications](#)
8. [CP Applications and Algorithms](#)
9. [CP Algorithms](#)
10. [CP Algorithms, Tensor Eigenvectors](#)
11. [CP Algorithms](#)
12. [Atomic Norms](#)
13. [Tensor Completion](#)
14. [Tensor Completion](#)
15. [Tensor Networks](#)
16. [Matrix Product States, Tensor Trains](#)
17. [Tensor Train Decompositions](#)
18. [Tucker Decompositions, Graphical Models](#)
19. [Nonnegative Matrix Factorization](#)
20. [Nonnegative Matrix and Tensor Factorization](#)
21. [Nonnegative Tensor Factorization and Total Positivity](#)
22. [Linear Non-Gaussian Acyclic Models](#)

7 Fast Algorithms and Integral Equation Methods

8 Compressed Sensing and Sparse Signal Processing

9 Solving Partial Differential Equations in Parallel on GPUs

1. [Julia Programming, PDE Solving, GPU Computing, and HPC](#)
2. [Finite-Difference Modeling of Diffusion, Waves, Advection, and Elliptic PDEs](#)
3. [Julia package management, environments, elliptic PDEs, and damped wave methods](#)
4. [Dual Time Stepping, Coupled PDEs, and Thermal Porous Convection](#)
5. [Implicit PDE Solvers, CPU Parallel Computing, Performance Metrics, and Memory Throughput](#)
6. [Julia Testing, Reference Tests, Continuous Integration, GitHub Actions, and Benchmarking](#)
7. [GPU Architecture, CUDA Kernel Programming, Memory Throughput, and Performance Optimization](#)
8. [Distributed Computing in Julia: MPI, Domain Decomposition, Scaling, and Parallel Efficiency](#)
9. [Code Documentation in Julia: Comments, Docstrings, Markdown, and Tooling](#)
10. [Advanced GPU Optimization with Shared Memory, Registers, and Performance Analysis](#)

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